MIL-O-85432B(AS)
31 AUGUST 1988
SUPERSEDING
MIL-O-85432A(AS)

MILITARY SPECIFICATION

OSCILLATOR, VOLTAGE CONTROLLED

The specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, New Jersey 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC: N/A

DISTRIBUTION STATEMENT A: Approved for public release, distribution is unlimited.

FSC 5865

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This specification establishes the performance test, manufacture, and acceptance requirements for voltage controlled oscillators (VCO) referred to herein as the oscillator.

1.2 Classification. Oscillators consist of the following types as specified in the contract (see 6.2.1).

a. Type I 4.0 to 8.0 gigahertz (GHz) in accordance with drawing 1356AS400

b. Type II 8.0 to 11.0 GHz in accordance with drawing 1356AS401

c. Type III 1.0 to 2.0 GHz in accordance with drawing 1356AS402

d. Type IV 2.0 to 4.0 GHz in accordance with drawing 1356AS403

e. Type V 11.0 to 18.0 GHz in accordance with drawing 1356AS404

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and (Repair Parts): Packaging of

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification and Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-794	Parts and Equipment, Procedures for Packaging, and Packing of
MIL-STD-810	Environmental Test Methods & Engineering Guidelines
MIL-STD-1695	Environments, Working, Minimum Standards for
MIL-STD-45662	Calibration Systems Requirements

HANDBOOKS

MIL-HDBK-781	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident. 30003)

1356AS400	Voltage Controlled Oscillator 4-8 GHz
1356AS401	Voltage Controlled Oscillator 8-11 GHz
1356AS402	Voltage Controlled Oscillator 1-2 GHz
1356AS403	Voltage Controlled Oscillator 2-4 GHz
1356AS404	Voltage Controlled Oscillator 11-18 GHz

(Copies of specifications, standards, handbooks and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Oscillator description. The oscillator is a continuous wave (CW), radio frequency (RF) generator whose output frequency is determined by the magnitude of a direct current (DC) tuning voltage input.

3.1.1 Components. The oscillator shall contain the following components:

a. Voltage controlled oscillator

b. Integral isolators

c. Proportionally controlled heaters

3.2 Characteristics.

3.2.1 Performance characteristics. The oscillator shall meet all the performance requirements specified in 3.2.1.1 through 3.2.2.5 when utilized in conjunction with other units of the AN/DLQ-3C(V) Countermeasures Set.

3.2.1.1 Frequency range. The frequency range for each type of oscillator shall be as specified in the contract (see 6.2.1).

3.2.1.2 Power output. The power output for the oscillator shall be as follows:

a.	Types I, II	Not less than +13 decibels above one milliwatt (dBm) and constant within 6 decibels (dB) across the frequency.
b.	Types III, IV	Not less than +17 dBm and constant within 6 dB across the frequency range.
c.	Type V	Not less than +10 dBm and constant within 6 dB across the frequency range.

3.2.1.3 Output impedance. The output impedance (Z) for all oscillators shall be nominally 50 ohms.

3.2.1.4 Spurious output. The spurious outputs (below carrier) shall not be less than 18 dB for harmonically related outputs and not less than 60 dB for nonharmonically related outputs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

3.2.1.5 Frequency pushing. The change in frequency over a range of ±0.5 volt (V) change in the +20 and -20 volts direct current (Vdc) voltages shall be not greater than ±0.1 percent per volt.

3.2.1.6 Frequency pulling. When operating into a voltage standing wave ratio (VSWR) of 2.5:1.0 which is varied through 360 electrical degrees, the oscillator frequency change shall be not greater than the following:

a.	Types I, II, and IV	±0.2 percent
b.	Type III	±1.0 percent
c.	Type V	±0.1 percent

3.2.1.7 Residual peak frequency modulation. The oscillator residual peak frequency modulation (FM) shall be not greater than the following:

a.	Type I	15 kilohertz (kHz)
b.	Type II	20 kHz
c.	Type III	5 kHz
d.	Type IV	10 kHz
e.	Type V	200 kHz

3.2.1.8 Modulation sensitivity. The bandwidth at the 3 dB points (modulation sensitivity) with a 15V peak-to-peak sawtooth waveform signal applied to the tuning ports at a 4 megahertz (MHz) rate from a 1500 ohm impedance driver shall be not less than the following:

a.	Types I, III, and IV	400 MHz
b.	Types II, V	800 MHz

3.2.1.9 Temperature drift. The temperature drift for the oscillator shall be as follows:

a.	Types I, II, V	Not greater than 30 parts per million (ppm) per degree celsius (^{0}C).
b.	Type III	Not greater than 100 ppm per ^{0}C.
c.	Type IV	Not greater than 80 ppm per ^{0}C.

3.2.2 Input requirements. The input requirements shall be as specified in 3.2.2.1 through 3.2.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

3.2.2.1 Input power. The oscillator shall utilize +20.0 ±0.5 and -20.0 ±0.5 Vdc for input power. The current drain for each voltage shall be not greater than the following:

a.	Types I, IV, and V	300 milliamperes (mA)
b.	Type II	150 mA
c.	Type III	420 mA

3.2.2.2 Heater power. The heater voltage shall be +28.0 ±2.0 Vdc and the current drain shall be not greater than 3.2 amperes (A) for all types.

3.2.2.3 Tuning voltage. The oscillator tuning voltage shall be within 0 to -60 Vdc and the current drain shall be not greater than 3.0 mA.

3.2.2.4 Output load mismatch. The oscillator shall neither be damaged nor its operation impaired, except for reduced power output, when operating into an open or short circuit. This mismatch applies to all load phase angles. Power so reflected shall be dissipated in an isolating device integral to the oscillator.

3.2.2.5 Voltage spikes and transients. The oscillator shall be in accordance with the requirements of table II and figure 10 of MIL-STD-704, and shall not be damaged by power line voltage spikes or transients of an amplitude up to 25V (including normal power supply voltage) and a duration of up to 100 milliseconds (ms).

3.2.3 Physical characteristics. Physical characteristics shall be as specified in 3.2.3.1 through 3.2.3.4.

3.2.3.1 Conformance to documents. The oscillator shall be in accordance with the applicable drawing as specified in the contract and shall meet all requirements specified herein (see 6.2.1).

3.2.3.2 External adjustments. The oscillator shall not have operating adjustments externally accessible.

3.2.3.3 Weight. The oscillator weight shall be not greater than 25 ounces (oz).

3.2.3.4 Baseplate flatness. The overall baseplate flatness shall be maintained within 0.010 inch.

3.3 Design and construction.

3.3.1 Materials. All materials used in the construction of the oscillator shall be in accordance with MIL-STD-454, Requirement 3, Flammable Materials; Requirement 4, Fungus-Inert Materials; Requirement 15, Ferrous Alloys, *Corrosion Resistance; and Requirement 16, Dissimilar Metals.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

3.3.2 Form factor. The outline dimensions and mounting configurations of the oscillator shall be as specified in the applicable drawing as specified in the contract (see 6.2.1).

3.3.3 Identification and marking. The nameplate shall be as specified in the applicable drawings as specified in the contract (see 6.2.1). Data for each oscillator type is as specified in table I. Interchangeable parts and assemblies shall be legibly marked in accordance with MIL-STD-130.

TABLE I. OSCILLATOR TYPE DATA.

Type	Drawing no.	Frequency	Assembly no.
I	1356AS400	4.0 to 8.0 GHz	1356AS208
II	1356AS401	8.0 to 11.0 GHz	1356AS207
III	1356AS402	1.0 to 2.0 GHz	1356AS303
IV	1356AS403	2.0 to 4.0 GHz	1356AS304
V	1356AS404	11.0 to 18.0 GHz	1356AS305

3.3.4 Interchangeability. The oscillator interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.3.5 Production processing. The oscillator shall be manufactured, inspected and tested under conditions and procedures meeting the following requirements specified in 3.3.5.1 through 3.3.5.4.

3.3.5.1 Special working environment. Adequate facilities shall be provided for the fabrication, assembly, and testing of the oscillator to be delivered in accordance with this specification. The facilities used in the manufacture, testing, and inspection of the oscillator shall be in accordance with MIL-STD-1695, Class C.

3.3.5.2 Inspection and test planning. The inspection program shall include the necessary planning for tests and inspections conducted during the entire phase of fabrication, processing, and assembly. The planning shall be based on a comprehensive study of the articles, fabrication and processing operations, methods of material integration, assembly and checkout, and the final test and inspection procedures. Inspections shall be established at points that will minimize delays resulting from deficiencies and, in all cases, shall be at or before the last point at which the acceptability of the operation or quality of the characteristics may be verified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

3.3.5.3 Process control procedures. Process control procedures shall be developed to supplement applicable process specifications and provide detailed performance and control methods. These procedures shall document the preparation; fabrication details; conditions to be maintained during each phase of the process; the methods of verifying the adequacy of processing materials, solutions, equipment and their associated control parameters, including statistical quality control plans where applicable; and the required records to indicate the results of such inspection and process verification.

3.3.5.4 Material control. Material controls shall ensure that only conforming materials and articles are used. Materials and articles not conforming to or not required for the operation involved shall be removed from the work operations. Positive action shall be taken to protect controlled processes or operations from contamination by residue from noncomforming materials and previous operations.

3.4 Reliability. Reliability testing shall be conducted on the first article oscillator. The oscillator shall have a mean-time-between-failure (MTBF) of not less than 100 hours.

3.4.1 Shelf life. The oscillator shall be designed to meet the performance requirements specified herein after a storage period of up to 5 years.

3.4.2 Maintainability. The oscillators are not characterized as being field maintainable at the organizational (O) or intermediate (I) levels of maintenance (see 6.3).

3.5 Environmental conditions. The oscillators shall meet all requirements as specified in MIL-E-5400 for class two equipment prior to, during, and after the following environmental conditions, separately or in any combination thereof. The oscillators shall not exhibit any temporary or permanent degradation as the result of exposure to any of the following:

a.	Temperature-altitude:	Barometric pressure reduced to the equivalent of 50,000 feet altitude and a temperature of -40^{0}C.
b.	Vibration:	Radom vibration at 10 to 2000 hertz (Hz) and $\pm$10 gravity units (g) root mean square (RMS). No vibration isolators shall be used.
c.	Shock:	Acceleration pulses at 12 g levels with a duration of 11 ms in all axes in any direction. No shock isolators shall be used.
d.	Humidity:	An atmosphere containing 95 percent or greater relative humidity at temperatures from +28 to $+71^{0}$C

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

e.	Storage temperature:	Temperatures ranging from -54^{0}C to $+71^{0}$C.
f.	Operating temperature:	Temperatures ranging from -40^{0}C to $+71^{0}$C.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.5 and 6.4).

3.6.1 First article sample. The first article sample shall be as specified in the contract or purchase order (see 6.2.1). The first article sample shall be manufactured using the same methods, materials, processes and procedures proposed for production.

3.7 Soldering. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5.

3.8 Workmanship. The oscillator, including all parts and accessories shall be constructed and finished in a manner that will assure compliance with all requirements specified herein. Unless otherwise specified herein, fabricating and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for conformance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5)

(1) Reliability tests (see 4.6)

b. Quality conformance inspections (see 4.7)

(1) Acceptance tests (see 4.7.1)

4.3 Inspection conditions. Unless otherwise specified herein, tests and inspections shall be conducted under the following conditions:

a.	Temperature (ambient):	+23 to $\pm 10^{0}$C
b.	Atmospheric pressure:	725 +50, -115 millimeter of mercury (28.5 +2.0, -4.5 inches of mercury)
c.	Humidity:	10 to 90 percent

4.4 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), the first article sample shall consist of one of each type of oscillator (see 1.2). When first article is required, first article inspection shall be conducted by the contractor. Approval of first article will be by the procuring activity after satisfactory completion of all tests. The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to the acceptance of the first article sample shall be at the risk of the contractor. Subsequent to first article approval, the contractor shall not change materials, processes, or procedures without prior approval of the procuring activity.

4.5 First article inspection. First article inspection shall consist of the examinations and tests of table II and shall be performed in the sequence specified.

4.5.1 First article failure analysis and corrective action. Failure of any oscillator to pass any of the first article examinations and tests specified herein shall be cause for rejection of the first article sample. Procuring activity approval is required before implementation of any corrective action in first article oscillators.

4.5.2 Post test. Post test shall consist of verification of the following requirements:

a. Frequency range (see 3.2.1.1)

b. Power output (see 3.2.1.2)

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection.

Examination or test	Requirement paragraphs	Method Paragraph First Article	Acceptance
1. Visual examinations			
a. Dimensions	3.3.2	4.9.1	4.9.1
b. Weight	3.2.3.3	4.9.1	4.9.1
c. Nameplate and markings	3.3.3	4.9.3	4.9.3
d. Interchangeability	3.3.4	4.9.4	
e. Baseplate flatness	3.2.3.4	4.9.1	4.9.1
f. Materials	3.3.1	4.9.2	
2. Performance tests			
a. Frequency range	3.2.1.1	4.10.1.1	4.10.1.1
b. Power output	3.2.1.2	4.10.1.1	4.10.1.1
c. Output impedance	3.2.1.3	4.10.1.2	
d. Spurious output	3.2.1.4	4.10.1.3	4.10.1.3
e. Frequency pushing	3.2.1.5	4.10.1.4	
f. Frequency pulling	3.2.1.6	4.10.1.5	
g. Residual peak frequency modulation	3.2.1.7	4.10.1.6	
h. Modulation sensitivity	3.2.1.8	4.10.1.7	4.10.1.7
i. Temperature drift	3.2.1.9	4.10.1.8	
j. Input power	3.2.2.1	4.10.1.9	
k. Heater power	3.2.2.2	4.10.1.9	
l. Tuning voltage	3.2.2.3	4.10.1.10	
m. Output load mismatch	3.2.2.4	4.10.1.11	
n. Voltage spike and transients	3.2.2.5	4.10.1.12	
3. Environmental tests			
a. Temperature-altitude	3.5	4.11.1.1	
b. Post-test		---	4.5.2
c. Low temperature	3.5	4.11.1.2	
d. Post test		---	4.5.2
e. High temperature	3.5	4.11.1.3	
f. Post test		---	4.5.2
g. Vibration		3.5	4.11.1.4
h. Post test		---	4.5.2
i. Shock	3.5	4.11.1.5	
j. Post test		---	4.5.2
k. Humidity		3.5	4.11.1.6
l. Post test		---	4.5.2
4. Reliability	3.4	4.6	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

c. Frequency pushing (see 3.2.1.5)

d. Frequency pulling (see 3.2.1.6)

e. Residual peak FM (see 3.2.1.7)

f. Modulation sensitivity (see 3.2.1.8)

g. Temperature drift (see 3.2.1.9)

4.6 Reliability test. Reliability test shall be performed on the first article sample. Test plan V-D of MIL-HDBK-781 shall be used to determine acceptance or rejection of the reliability test sample. (Theta) shall be 100 hours. All time accumulated in environmental testing (see 4.11) shall be counted in determining whether an accept or reject condition exists.

4.7 Quality conformance.

4.7.1 Acceptance test. Each oscillator offered for acceptance shall be subjected to the tests of table II in the sequence specified therein. Any oscillator that fails to meet the specified requirements shall be rejected.

4.8 Test documentation and test equipment.

4.8.1 Test and inspection records. When specified in the contract or purchase order, test and inspection records shall be available for the Government representative. The test and inspection records shall include objective quality evidence in accordance with MIL-STD-109 that the components used as part of the oscillator meet the applicable requirements (see 6.2.2).

4.8.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance for the variable to be measured. In some cases, where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within 5 percent of that requirement.

4.8.2.1 Maintenance and calibration. All test equipment shall be maintained and calibrated in accordance with MIL-STD-45662.

4.9 Visual and physical examination.

4.9.1 Physical examination. The oscillator shall be examined for conformance with 3.2.3 and 3.3.2.

4.9.2 Materials. The oscillator shall be physically and functionally examined for conformance with 3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

4.9.3 Nameplate and markings. The oscillator shall be visually examined for conformance with 3.3.3.

4.9.4 Interchangeability. The oscillator shall be examined for conformance with 3.3.4.

4.9.5 Workmanship. The oscillator shall be visually examined for conformance with 3.7 and 3.8.

4.10 Performance requirements.

4.10.1 Performance tests. Performance requirements specified in 3.2.1 shall be verified in accordance with the approved test procedures specified in 4.8.1 and the guidelines provided in 4.10.1.1 through 4.10.1.12.

4.10.1.1 Frequency range and power output. The performance requirements specified in 3.2.1.1 and 3.2.1.2 shall be verified in accordance with the following frequency range and power output test procedure. Applicable test equipment shall be connected to the oscillator under test as shown on figure 1. The test arrangement shall be calibrated for loss through the pad and coupler. The tuning voltage shall be adjusted to obtain minimum output frequency and increased in 200 MHz. The output frequency shall then be slowly varied from maximum to minimum and output power levels and differences there of shall likewise be noted and recorded.

4.10.1.2 Output impedance and VSWR. Output impedance and VSWR are designed considerations requiring only contractor certification of compliance and need not be verified.

4.10.1.3 Spurious output and suppression. The performance requirements specified in 3.2.1.4 shall be verified in accordance with the following spurious output suppression guidelines. Applicable test equipment shall be connected to the oscillator under test as shown on figure 2.

a. The oscillator turning voltage shall be adjusted to the minimum frequency point on the band.

b. The spectrum analyzer shall be adjusted to twice the output frequency of the oscillator under test.

c. The tuning voltage shall be slowly increased while tracking the harmonic signal on the analyzer. The worst case spurious suppression shall be expressed in dB on the test data sheet (see 4.8.2).

d. The spectrum analyzer shall be readjusted to one-half (1/2) of the output frequency of the oscillator under test.

e. Steps a and c shall be repeated.

f. The spectrum analyzer shall be readjusted to three-halves (3/2) of the output frequency of the oscillator under test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

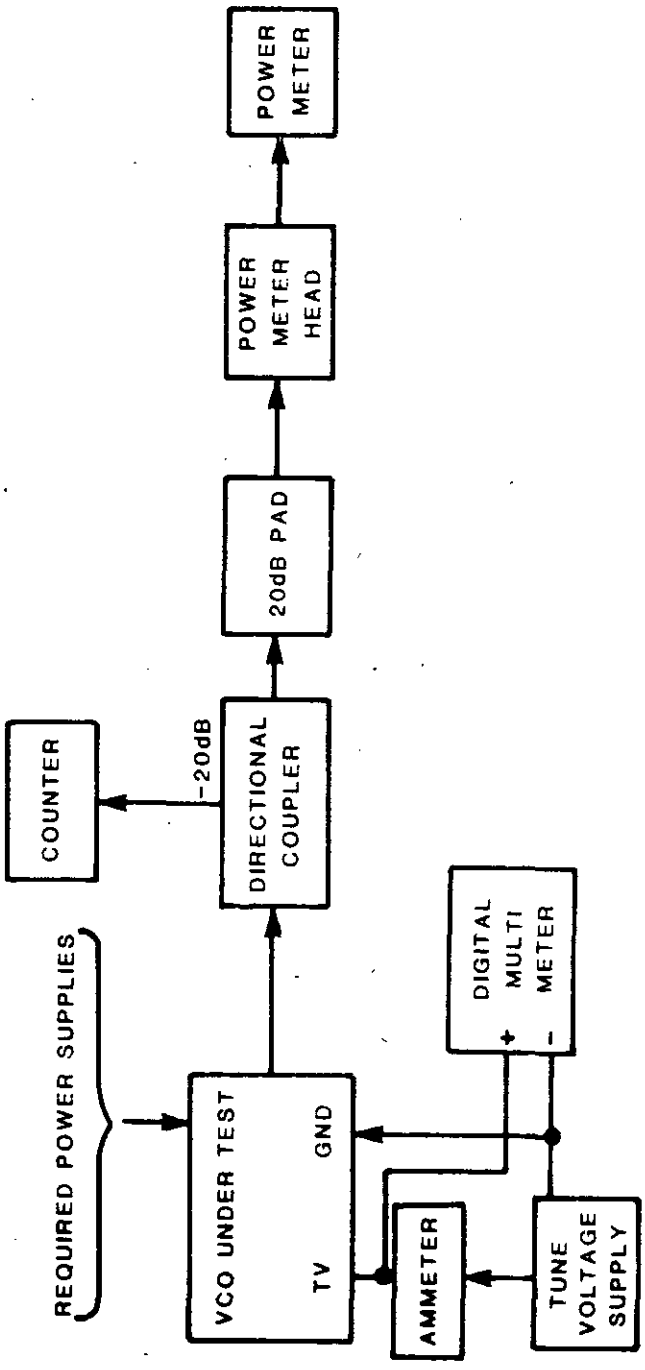


FIGURE 1. Connection of test equipment for frequency range and power output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

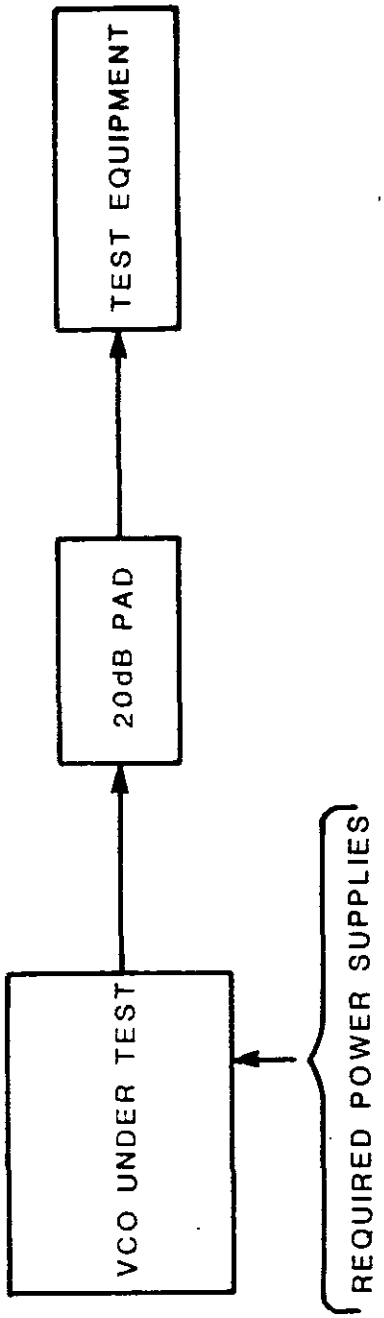


FIGURE 2. Connection of test equipment for spurious output suppression.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

g. Steps a and c shall be repeated.

h. Step a shall be repeated.

i. The oscillator tuning voltage shall be increased slowly to its maximum frequency while monitoring the entire oscillator tuning range to verify that all spurious outputs shall be in accordance with 3.2.1.4.

4.10.1.4 Frequency pushing. The performance requirements specified in 3.2.1.5 shall be verified in accordance with the following frequency pushing guidelines. Applicable test equipment shall be connected to the oscillator under test as shown on figure 2.

a. The tuning voltage of the oscillator under test shall be adjusted to the minimum output frequency point on the band and the signal shall be identified.

b. An EIP Model 351C, or equal, frequency counter may be used in lieu of the spectrum analyzer and, if elected, step b may be disregarded. The spectrum analyzer shall be adjusted to the following conditions:

(1)	IF band	1 kHz
(2)	Scan width	1 MHz/division
(3)	Scan time/division	Adjust until calibration light extinguishes

c. The positive (+) voltage supply shall be varied as specified in 3.2.1.5. The peak-to-peak frequency deviation shall be measured and recorded on the test data sheet (see 4.8.2).

d. A negative (-) voltage supply shall be substituted for the (+) voltage supply and step c shall be repeated.

e. Steps b, c, and d shall be repeated for the middle and maximum output frequency points.

4.10.1.5 Frequency pulling. The performance requirements specified in 3.2.1.6 shall be verified with the following frequency pulling guidelines. Applicable test equipment shall be connected to the oscillator under test as shown on figure 3.

a. The tuning voltage of the oscillator under test shall be adjusted to the minimum output frequency point of the band and the signal shall be identified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

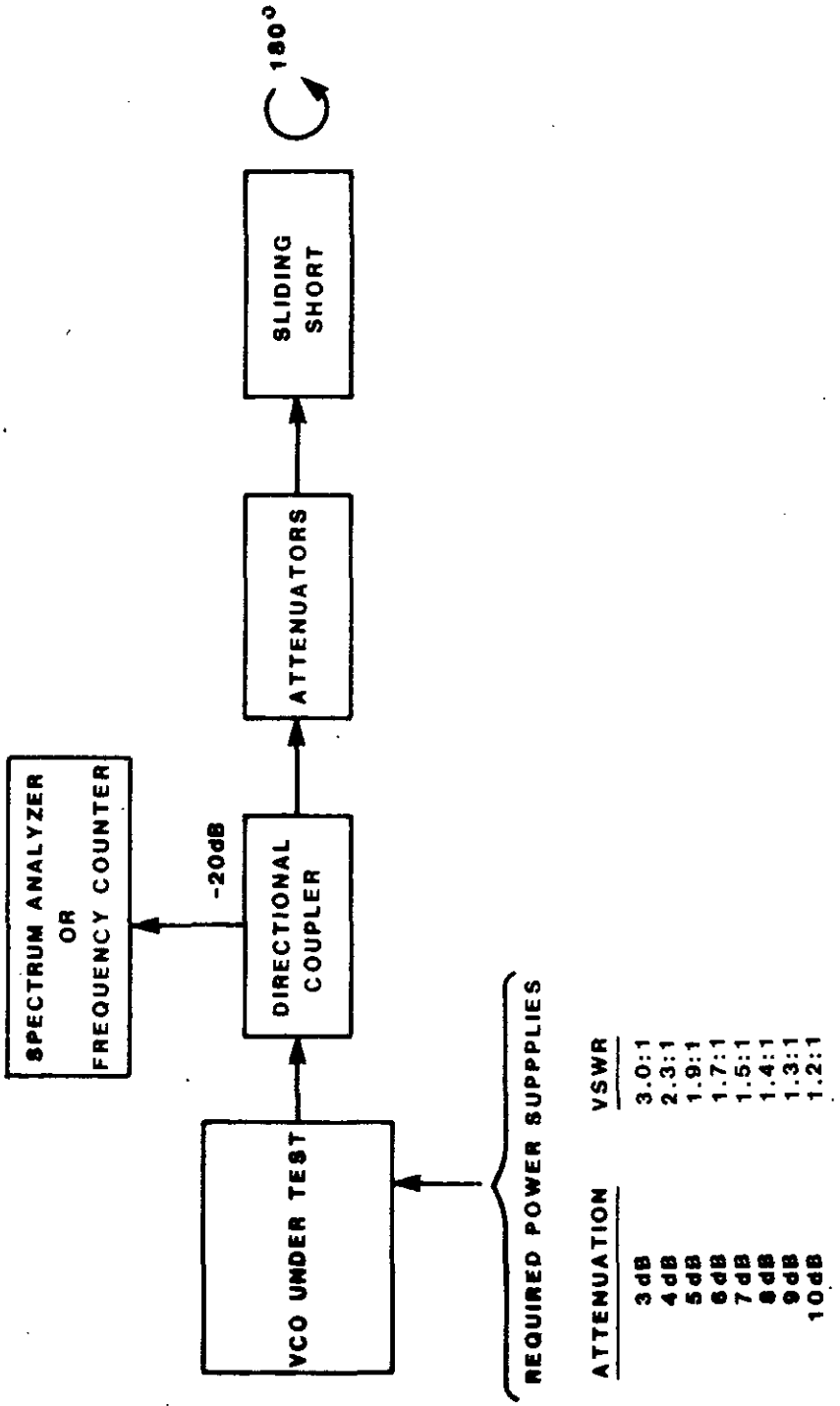


FIGURE 3. Connection of test equipment for frequency pulling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

b. An EIP Model 351C, or equal, frequency counter may be used in lieu of the spectrum analyzer and, if elected, step b may be disregarded. The spectrum analyzer shall be adjusted to the following conditions:

(1)	IF band	1 kHz
(2)	Scan width	Less than one-half of the peak-to-peak pulling per division specified in 3.2.1.6.

c. The sliding short mismatch shall be rotated through all possible phases by movement through its entire range. The peak-to-peak frequency deviation shall be measured and recorded on a suitable test data sheet (see 4.8.2).

d. Steps b and c shall be repeated for the middle and maximum output frequency points.

4.10.1.6 Residual peak FM. The performance requirements specified in 3.2.1.7 shall be verified with the following residual peak FM guidelines. Applicable test equipment shall be connected to the oscillator under test as shown on figure 2.

a. The tuning voltage of the oscillator under test shall be adjusted to the minimum output frequency point of the band and the signal shall be identified.

b. The spectrum analyzer shall be adjusted to the following conditions:

(1)	IF band	1 kHz
(2)	Scan width	Equal to or less than 20 kHz/division
(3)	Scan time/division	Equal to or less than 20 ppm/Div T (sec/div) = Scan Width (Hz/div) 20 x 10 (F) scan
(4)	Calibration light	Off

c. The width of the output signal shall be measured at the 3 dB points below the peak carrier signal as shown on figure 4.

d. Steps a, b, and c shall be repeated for the middle and maximum output frequency points.

4.10.1.7 Modulation sensitivity. The performance requirements specified in 3.2.1.8 shall be verified with the following modulation sensitivity guidelines. Applicable test equipment shall be connected to the oscillator under test as shown on figure 5.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

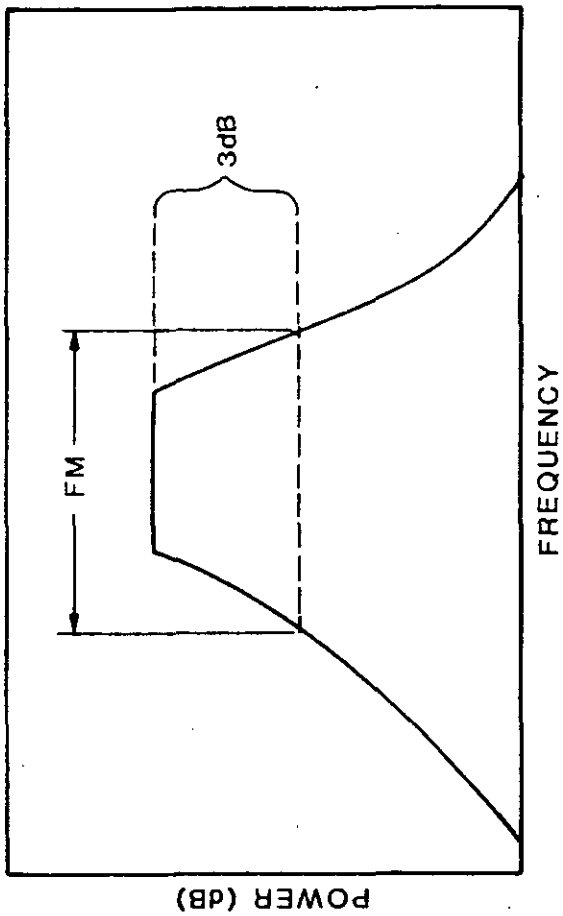


FIGURE 4. Measurement of width of output signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

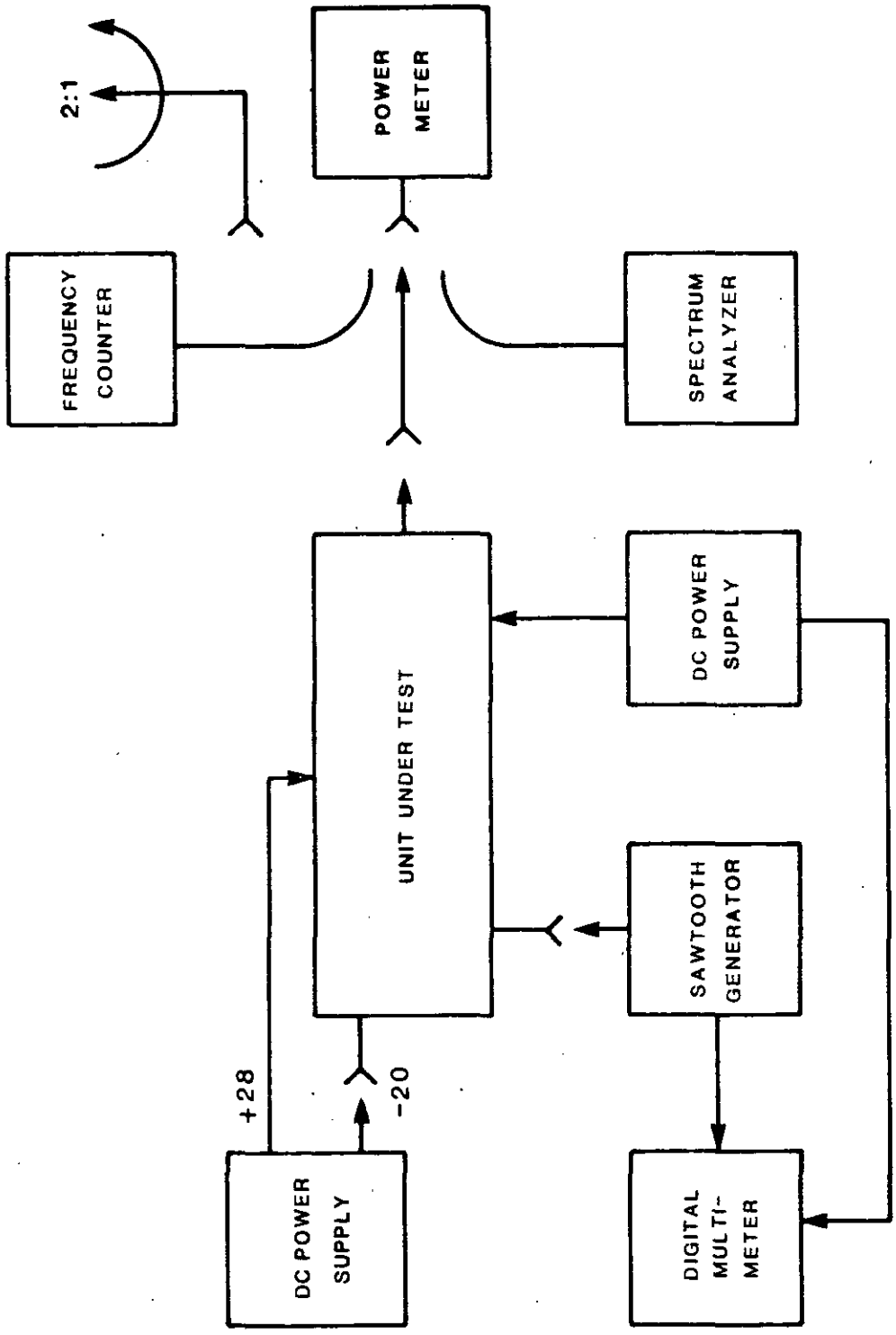


FIGURE 5. Connection of test equipment for modulation sensitivity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

a. The tuning voltage of the oscillator under test shall be adjusted to the minimum output frequency point of the band and the signal shall be identified.

b. The modulation rate of the sawtooth waveform generator shall be adjusted to 100 kHz. The output impedance of the driver shall be verified 1500 ohms.

c. The amplitude of the sawtooth waveform generator output shall be adjusted to 15 V peak-to-peak.

d. The dispersion of the spectrum analyzer shall be nominally adjusted to 200 MHz/per centimeter.

e. The carrier width of the output signal shall be measured at the 3 dB points below peak carrier signal as shown on figure 4.

f. Steps a through e shall be repeated for the middle and maximum frequency points.

4.10.1.8 Temperature drift. The performance requirements specified in 3.2.1.9 shall be verified with the following temperature drift guidelines. The oscillator under test shall be installed in a temperature chamber with adequate heat sink compound between the oscillator and chamber baseplate. The following steps apply, however, the order of steps b and f may be reversed for the sake of convenience.

a. A frequency counter shall be connected to the oscillator under test as shown on figure 5.

b. The chamber temperature measured at the baseplate shall be adjusted to the low temperature specified in 3.5 herein and stabilized for a period of not less than 10 minutes.

c. The tuning voltage shall be adjusted to zero volts.

d. The tuning voltage shall be increased in 1 V steps up to the maximum output frequency point. The output frequency at each step shall be recorded and temperature drift in terms of ppm shall be computed.

e. The steady-state heater current shall be recorded on the test data sheet utilized in step d.

f. The chamber temperature measured at the baseplate shall be readjusted to the high temperature specified in 3.5 and stabilized for a period of not less than 10 minutes.

g. Steps c, d, and e shall be repeated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

4.10.1.9 Input and heater power. These powers shall be measured under ambient laboratory conditions.

4.10.1.10 Tuning voltage. The performance requirements specified in 3.2.2.3 shall be verified with the following tuning voltage guidelines. Applicable test equipment shall be connected to the oscillator under test as shown on figure 2.

a. The oscillator shall be tuned to the minimum frequency point on the band, and the voltage and current shall meet the requirements of 3.2.2.3.

b. The oscillator shall be tuned to the maximum frequency point on the band, and the voltage and current shall meet the requirements of 3.2.2.3.

4.10.1.11 Output load mismatch. The oscillator shall be operated into an open-end short circuit to verify compliance with 3.2.2.4.

4.10.1.12 Voltage spike and transients. The oscillator shall be tested in accordance with the approved test procedures of 4.8.1 to verify compliance with 3.2.2.5.

4.11 Environmental tests. Environmental tests shall be as specified in 4.11.1 through 4.11.1.6.

4.11.1 Environmental conditions. The environmental conditions shall be as specified in 3.5.

4.11.1.1 Temperature-altitude. The oscillator shall be tested as specified in the temperature-altitude test in MIL-T-5422 for class 2 equipment except as modified by 3.5. Following completion of step 14 of the MIL-T-5422 temperature-altitude test, the tests specified in 4.5.2 shall be conducted.

4.11.1.2 Low temperature. The oscillator shall be tested in accordance with MIL-STD-810, Method 502.2, Procedure II. The test item configuration shall be that of a realistically deployed, unprotected oscillator. The test temperature shall be -40^{0}C. Step six of Method 502.2, II-2.2 shall consist of the post tests specified in 4.5.2 herein. Step one of Method 502.2, II-3.2 shall be 24 hours. Steps four and 10 of II-3.2 shall be the post tests specified in 4.5.2 herein. Step six shall be omitted. Step seven shall consist of external power application for a period of not less than 2 hours.

4.11.1.3 High temperature. The oscillator shall be tested in accordance with MIL-STD-810, Method 501.2, Procedure II. Operating temperature shall be $+71^{0}$C. The test item configuration shall be that of a realistically deployed, unprotected oscillator. Relative humidity during the test shall be not greater that 15 percent. Temperature stabilization of the test item shall consist of external power application for a period of not less than two hours while the operating temperature is maintained. Operational checkout during all phases shall consist of the post tests specified in 4.5.2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

4.11.1.4 Vibration. The oscillator shall be tested in accordance with MIL-STD-810, Method 514.3, Procedure I, as specified for category 7B equipment. Test item configuration shall be that of a realistically deployed, unprotected oscillator. Vibration isolators shall not be used. The oscillator shall be vibrated along three axes, one axis at a time, for not less than one hour each per cycle. The oscillator shall be operated continuously, as if it were in operational use during test. Vibration spectrum and intensity shall be as shown on figure 6. The operational and functional check specified in II-2.1, step seven and II-3.1, step seven, shall consist of the tests specified in 4.5.2 herein.

4.11.1.5 Shock. The oscillator shall be tested in accordance with MIL-T-5422, part II. A crash safety test is not required. Upon completion of the basic design test, the tests specified in 4.5.2 herein shall be conducted.

4.11.1.6 Humidity. The oscillator shall be tested in accordance with the humidity test as specified in MIL-T-5422 except as modified by 3.5 herein. Step six shall consist of the post tests specified in 4.5.2 herein.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation, packaging, and packing shall be as specified in MIL-E-17555 and in accordance with MIL-STD-794, level C. The method of preservation for level C shall be determined as specified in the selection chart in appendix D of MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The oscillator unit is intended for use as a component of an electronic countermeasures set installed in aerial targets and piloted aircraft. The oscillator is of five distinct types which differ in physical size and electrical parameters.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type, frequency range, and drawing number of oscillator (see 1.2)

c. Requirements for and number of first article samples (see 3.6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

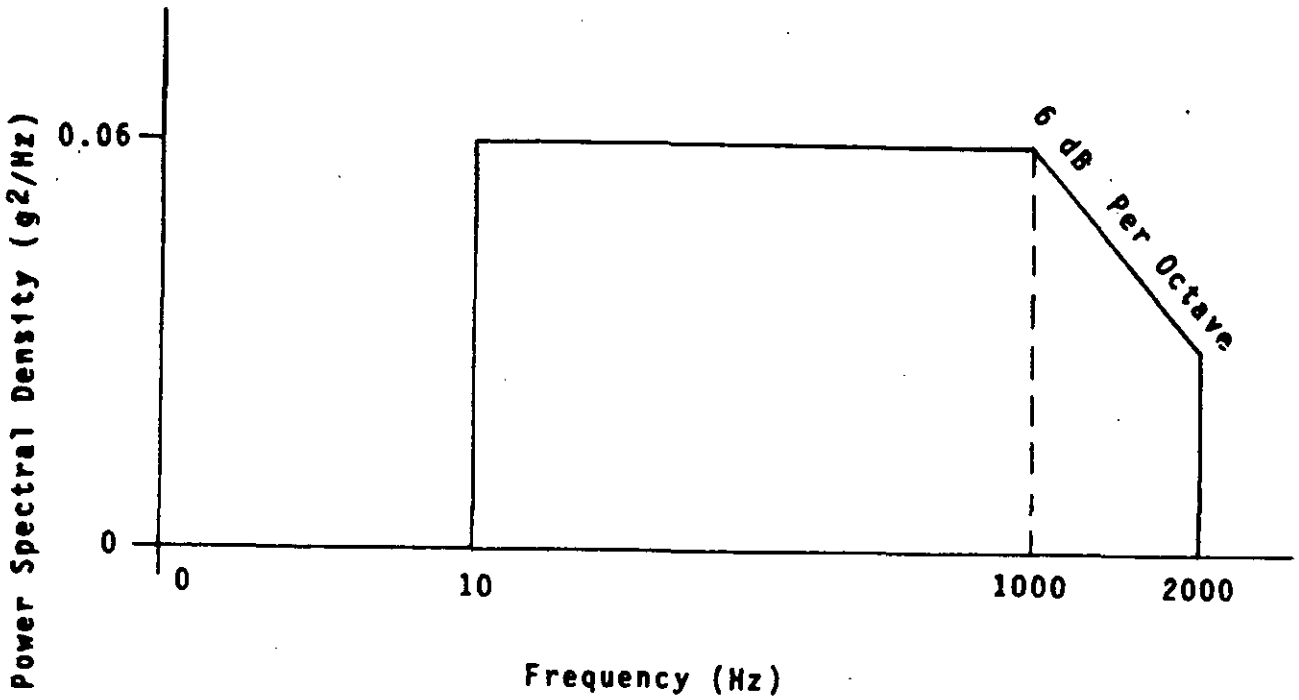


FIGURE 6. Vibration test level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

d. Responsibility for inspection (see 4.1).

e. Inspection facilities if different (see 4.1)

f. Level of preservation, packaging, and packing if different than specified (see 5.1).

g. Special markings, if required (see 5.2.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.
4.4	Engineering Change Proposals Requests for Deviations and Waivers (Long Form)	DI-E-2037
4.5	Report, First Article (Preproduction) Test	UDI-T-21349
4.5.1	Report, Failure/Malfunction	UDI-T-23724
4.6	Plan, Reliability Test	DI-RELI-80250
4.6	Procedures, Reliability Test	DI-RELI-80251
4.6	Reports, Reliability Test	DI-RELI-80252
4.6	Failed Item Analysis Report	DI-RELI-80253
4.7.1	Report, Failure	DI-R-21598
4.8.1	Test Plan	DI-T-5204
4.8.1	Procedures, Test	UDI-T-21347
4.8.1	Records, Inspection and Test	UDI-T-22706
4.8.2.1	Calibration System Description	DI-R-7064

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of DIDs required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

6.3 Definitions.

6.3.1 Organizational maintenance. All maintenance performed at or in the aircraft or RPV by the using organization employing only those skills, tools, support equipment, publications, procedures, and techniques planned for service use when deployed.

6.3.2 Intermediate maintenance. All maintenance other than organizational maintenance performed for direct support of the using activity, employing only skills, tools, and support equipment planned for normal service use on board an aircraft carrier or at a designated intermediate maintenance facility ashore.

6.4 First article. When first article inspection is required the item should be a first article sample. The first article should consist of one oscillator. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (keyword) listing.

Oscillator
Voltage Controlled
Countermeasures Set
AN/DLQ-3C(V).

Preparing Activity
Navy - AS
(Project 5865-N076)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-85432B(AS)	OSCILLATOR, VOLTAGE CONTROLLED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil – Downloaded: 2026-10-23T10:49Z
Check the source to verify that this is the current version before use.